Technology Stack

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

OptiCrop uses Python, Flask, HTML, CSS, JavaScript, a crop recommendation dataset, and GitHub to build an efficient and user-friendly crop recommendation system.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	HTML, CSS, JavaScript	Creates a responsive and user-friendly interface for farmers to enter soil and weather details and view crop recommendations.

2	**Backend / Server-Side**)	Python, Flask	Processes user input, loads the trained machine learning model, and generates crop recommendations.
3	**Database / Data Storage**	Crop Recommendation Dataset (CSV)	Stores agricultural data used for training and testing the machine learning model.
4	**Cloud / Hosting / Deployment**	Local System / Flask Development Server	Runs and tests the application during development with simple deployment.

5	**Version Control & CI/CD**	Git, GitHub	Tracks code changes, supports collaboration, and maintains project versions.
6	**Third-Party APIs / Other Tools**	Scikit-learn, Pandas, NumPy, Matplotlib	Used for data preprocessing, machine learning model training, prediction, and data visualization.